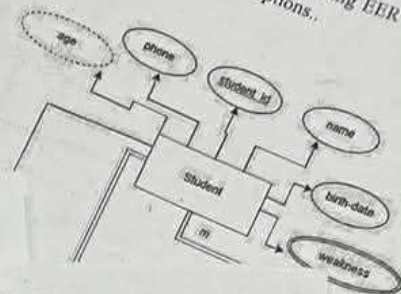


- 1) Draw a Schema
- 2) Describe the DBMS Language
- 3) What is Database and its components
- 4) Difference between

Rail
Train



F2N

BRAC University
Department of Computer Science and Engineering

Examination : Makeup Midterm
Duration : 75 Minutes

CSE370 : Database Systems
Set M

Semester : Spring 2025
Full Marks : 30

Answer All Three Questions On The Answer Script.
Understanding the Question is a Part of the Examination.
Do Not Ask Any Questions and Answer to the Best of Your Understanding.

Full Name: Ma Abu Taleb Forhad ID: 29191227

1

CO1

(a) Imagine you are a database administrator for an e-commerce platform, tasked with improving the database structure without affecting the applications. In one scenario, you need to add a new column to the product details table for the discount percentage, ensuring no impact on how the data is displayed or used by the application. In another, you must reorganize the internal file structure and create new indexes to improve performance, while maintaining the existing conceptual schema and ensuring users' interactions with the data remain unchanged.

Identify the type of data independence being described in each scenario and briefly explain your reasoning.

(b) Refer to the table below to answer the following questions:

Book					Loan					Member			
BookID	Title	Author	Genre	Year	LoanID	BookID	MemberID	DateBorrowed	DueDate	MemberID	Name	Email	Age
B101	The Lord of the Rings	J.R.R. Tolkien	Fantasy	1954	L101	B101	M101	2023-11-15	2023-11-29	M101	Alice	alice@gmail.com	25
B102	To Kill a Mockingbird	Harper Lee	Fiction	1960	L102	B102	M102	2023-12-01	2023-12-15	M102	Bob	bob@yahoo.com	22
B103	The Hitchhiker	Douglas Adams	Science Fiction	1979	L103	B103	M103	2023-11-21	2023-12-04	M103	Charlie	charlie@gmail.com	30
B104	Nineteen Eighty-Four	George Orwell	Dystopian	1949	L104	B104	M101	2023-12-10	2023-12-24	M104	David	david@yahoo.com	27

L105 B102

(i) If <'L105', 'B202', 105, '2023-10-19', 'Jan 4 2024'> is inserted into the Loan table, identify all constraints which will be violated by this operation and briefly explain your answer.

(ii) If you delete the book with BookID = 'B102' from the Book Table, identify all constraints which will be violated by this operation and explain how you will solve it.

2

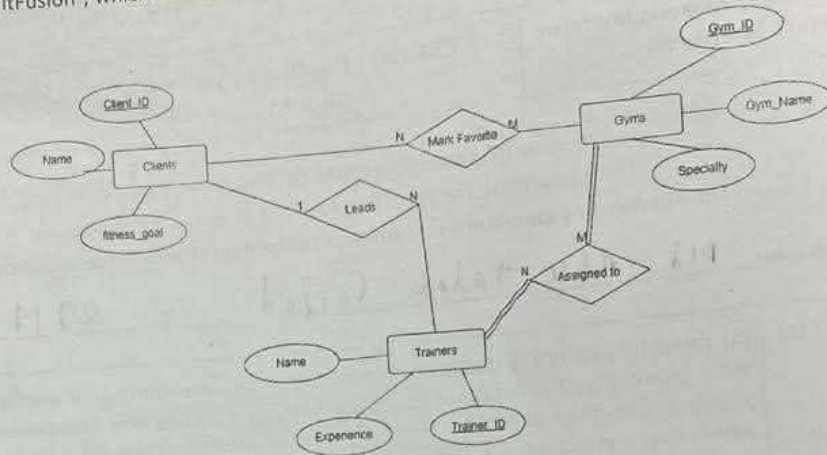
4

2

2

CO2

You are provided with a basic ER diagram for a wellness and training platform called "FitFusion", which includes Clients, Trainers, and Gyms.



Due to updated features and business rules, you must now revise the diagram. The revised requirements are as follows:

- Gyms in "FitFusion" do not have a Gym_ID anymore. Instead, they are uniquely identified by their Gym_Website. The website embeds the Gym_Name attribute, but Gym_Name can be shared by multiple gyms with different Specialties (e.g., CrossFit, Yoga, Powerlifting). Together, Gym_Name and Specialty uniquely identify a Gym.
- Clients are categorized into two types: Basic Clients and Premium Clients. All Clients can add Workout_Notes for their training sessions. A Workout_Note cannot exist independently without a Client. Each Workout_Note includes: session_address, note_number, and trainer_instruction. For every Client, note_number starts from 1.
- Basic Clients have an attribute Min_Session_Count. Premium Clients are further divided into Elite and Pro tiers. A Client cannot belong to both tiers. Elite Clients can set multiple Recovery Stations, which are locations where they can request post-workout recovery services (e.g., ice bath, massage).
- Trainers are the personnel who guide clients. Before becoming a certified Trainer, a person must undergo guidance from a Master Trainer. A Master Trainer can mentor one or more Trainers. Every new Trainer must have exactly one Master Trainer. Each mentorship session generates multiple Mentoring_Records, which include Feedback and Duration.
- Sessions represent logged interactions in the system. Each Session has a unique Session_ID, Total_Hours, and Session_Type. Clients can either attend sessions individually or as part of a Group. Trainers facilitate the session. For

		group sessions, multiple Trainers may conduct them jointly. Each session has a Session_Fee, calculated from Num_Trainers and a Base_Fee. Only Premium Clients can give <u>Feedback on their Sessions</u>. Their <u>feedback comment</u> is saved for each session they attend. (a) Update the existing ER diagram and construct a new Enhanced ER (EER) diagram incorporating the new requirements. Do not ignore any part of the original diagram. Use only the entities, attributes, relationships mentioned above. Do not assume any additional attributes, entities, relationships, key, multivalued, or composite attributes other than those indicated in the question. (b) State and explain the following questions with only one sentence: (i) Can a Gym be marked as Favourite by more than one Client? (ii) "All Trainers must be assigned to a Gym according to the ER diagram" – Do you agree or disagree? Justify your stance.	13
3	CO2	Design an Entity-Relationship (ER) Diagram for creating a DVD Rental Shop Database. Your database will store information about <u>DVDs</u>, <u>customers</u>, <u>rentals</u>, <u>genres</u>, <u>staff</u>, <u>late returns</u>, and more. Please note that the mentioned information may not all be represented as entities, some may be represented as relationships or attributes, depending on your design choices and general logic. You may design the ER diagram as you wish and include additional information, but your design must be logically accurate and should fulfill the following conditions: - There should be at least 3 regular entities and 1 weak entity. - There should be exactly 1 recursive relationship (it cannot be a mentorship/supervision/training or similar relationship) with its own attribute. - There should be at least 2 binary relationships, at least one of which should be a M:N relationship. - There should be one ternary relationship with total participation from all related entities and proper cardinality ratio. - There should be one composite-multivalued attribute.	7

BRAC University
Department of Computer Science and Engineering

Examination : Midterm
 Duration : 75 Minutes

CSE370 : Database Systems
Set B

Semester : Summer 2025
 Full Marks : 30

Answer **All Three** Questions On The Answer Script.

Understanding the Question is a Part of the Examination.

Do Not Ask Any Questions and Answer to the Best of Your Understanding.

1

(a) You work for an online bookstore that stores book prices as a float data type in its database. To ensure better precision in financial transactions, the company decides to change the data type from float to decimal(10,2) in the backend. Book prices are displayed on the website in the same format as before. **Identify** the type of data independence this scenario illustrates. Briefly **explain** your answer.

(b) Refer to the figure below to answer the following questions.

Borrower			Book			
<u>Borrower_ID</u>	Name	Book_ID	<u>Book_ID</u>	Title	Author	Pages
101	Arif	1	1	1984	George Orwell	328
102	Farhan	3	2	The Hobbit	J.R.R. Tolkien	310
			3	A Brief History of Time	Stephen Hawking	256

Figure: A

Suppose that each of the following operations (i, ii) is applied directly to the database state shown in the above figure (Figure A). Primary keys are underlined, foreign keys are shown using an arrow. *Consider each operation independent of each other.*

Identify all constraints violated by each operation and **explain** how the constraints can be enforced.

(i) INSERT INTO Book VALUES (null, 'Just Kids', 'Patti Smith', -270);

(ii) DELETE FROM Book where Title = '1984';

2

CO2

You are a database consultant hired by the BRACU Research Center (BRC). Their current system for tracking projects is outdated and unable to handle the complexity of their data. You are tasked with updating the existing database design given below.

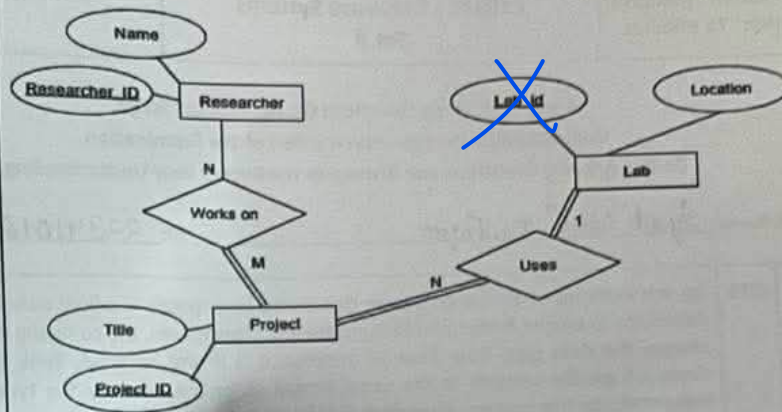


Figure: B

You have to update the above design (Figure B) according to the latest requirements given below:

- For all researchers, BRC must store their multiple academic_degrees and a list of all their publications, where each publication includes a Title and Year_of_Publication. The total number_of_projects for any researcher is shown, calculated by counting the current projects they are working on.
- Each lab is now uniquely identified by the combination of floorNumber and SerialNumber, instead of a single Lab_id.
- Researchers are now classified as either a Principal_Investigator (PI), having a Grant_Code, or a Research_Assistant (RA), having a Skill_Certification. No other type of researchers exists other than PI or RA. All junior researchers are supervised by exactly one senior researcher.
- The existing Project entity must be updated accordingly: the key attribute is now called Project_Code, a StartDate, and a Budget is added. Every project must be categorized as either Fundamental, having a Research_Question, or Applied, having an Industry_Partner.
- All Principal_Investigators have more than one Research_Grant. A Research_Grant cannot exist on its own; it must always be associated with the specific Principal_Investigator who secured it. A Research_Grant's Grant_Number is only unique for a given PI (thus different PIs may have the same Grant_Number). Each grant also has a Funding_Amount and Funding_Source.

		✓ The Works_On relationship between researchers and projects must also be modified to include a Role attribute, recording the specific role a researcher plays in each project. ✓ Every project must be managed by exactly one department. Each department has a unique name, and buildingNumber. Additionally, the BRC must track all Equipment, which has a unique Model_Number, and Manufacturer. A project may require multiple equipment, but the equipment can be used by only one project at a time. (a) Update the existing ER diagram (Figure B) to construct an EER diagram according to the above requirements. <i>Unchanged portions of the existing diagram must also be shown.</i> Do not assume any additional attributes, entities, relationships, keys, multivalued, or composite attributes other than those indicated in the question. (b) State and explain the following questions with only one sentence: (i) Does every Researcher work on a project according to the given ER diagram? (ii) "Each Project uses more than one Lab" - do you support or refute this statement?	13
3	CO2	Design an Entity-Relationship (ER) Diagram for a Music Concert Management System. Your database should include information about <u>artists</u>, <u>events</u>, <u>venues</u>, <u>visitors</u>, <u>ticketing</u>, <u>sponsors</u>, etc. Please note that all information may not be represented as entities; some may be represented as relationships/attributes, depending on your needs and general logic. You may design the ER as you wish and include additional information, but it must be logically accurate and should fulfill the following conditions: ✓ a. There should be at least 4 regular entities. ✓ b. There should be exactly 1 recursive relationship. ✓ c. There should be at least 2 binary relationships, at least one of which should be a 1:N relationship. ✓ d. There should be one derived attribute (except "age"). Explain in one sentence how your derived attribute will be calculated.	7